BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

LESSON 272 · Database and SQL retrieval challenge

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. What is a primary key?

Q2. What does INNER JOIN return?

Q3. What is a foreign key?

Q4. A 3×2 zero-indexed array: specify row index range, column range and cell count.

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I Do · Read new knowledge

Key vocabulary

primary key — Unique row identifier.

foreign key — Related table key reference.

INNER JOIN — Returns matching joined rows.

WHERE — Filters query results.

New learning

Understand — Use a primary key for each Student row and a matching foreign key in Score.

Trace — INNER JOIN produces one result for each matching StudentID; Cleo has no matching Score and is absent.

Evaluate — A WHERE clause can filter matched records; forgetting the join condition can produce unwanted combinations.

Worked example

Match Tutor and Class records on TutorID.

1. Predict first — Tutor 1 matches two class rows

2. Model and check — Tutor 2 has no class and is excluded from INNER JOIN.

We Do · Practise with guidance

Student IDs 4,5,6 but Mark IDs 4,6. What appears after INNER JOIN?

```
Student(4,Jo),(5,Rae),(6,Sol)
Mark(4,60),(6,90)
→ Jo 60; Sol 90
```

Predict first

IDs 4 and 6 match

Model and check

ID5 is absent from joined result.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

Define primary key, foreign key, INNER JOIN and WHERE. [4]

Question 2

Write an INNER JOIN for Student and Score above selecting Name and Mark; give output. [6]

Question 3

Explain why omitting a join condition may produce misleading results. [6]

Exit Ticket · Answer before checking

INNER JOIN automatically includes rows without a key match.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. Uniquely identifies a row.

Q2. Matched records according to join condition.

Q3. Column referencing a key in a related table.

Q4. Rows 0–2, columns 0–1, total six cells.

You Do pages · Independent questions

Q1. Unique ID [1], key reference [1], matching records [1], filtering predicate [1].

Q2. SELECT Name,Mark [1] FROM Student [1] INNER JOIN Score [1] ON Student.StudentID=Score.StudentID [1]; Ada75 and Ben82 [2].

Q3. Cartesian product [2], unconnected pairs [1], duplicated counts [1]; explicit join keys [1], check cardinality [1].

Exit Ticket · Model answer

False. Such rows are excluded.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: